Introduction

This is Applied Research for a given domain related project work. This will be a technical report, which uses domain related tools to experiment, test, evaluate, document and report back to the management or board.

Applied research is meant to reduce the time to market for a business objective by way of using research methods in a commercial setup. This helps the full time employers to work in a research environment and apply the learning for improving the products and services offered by the enterprise

This report has various sections that can be tailored to the unique needs of the desired outcome like improved product quality, service, better productivity etc.

Common sections

These are fundamentally same across industries. The objective and outcome is clear unlike the fundamental research that is based on hypothesis. Please refer to Applied Research common methods from your public libraries and search engines and populate this section. These can include scope of work, program and project management, budget, grants management, training, IT assets, engineering setup, data center, cloud and digital setup, licenses etc

Agricultural domain sections

How the common sections are extended to the agricultural domain, fill in refine, details the common sections as a separate section

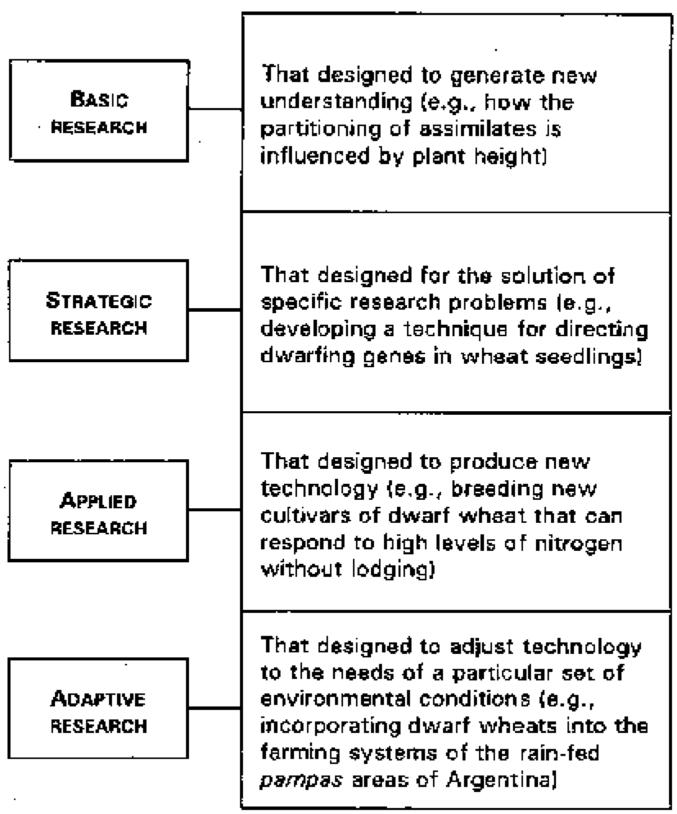

Fig 1: Research types in the agricultural industry [1]

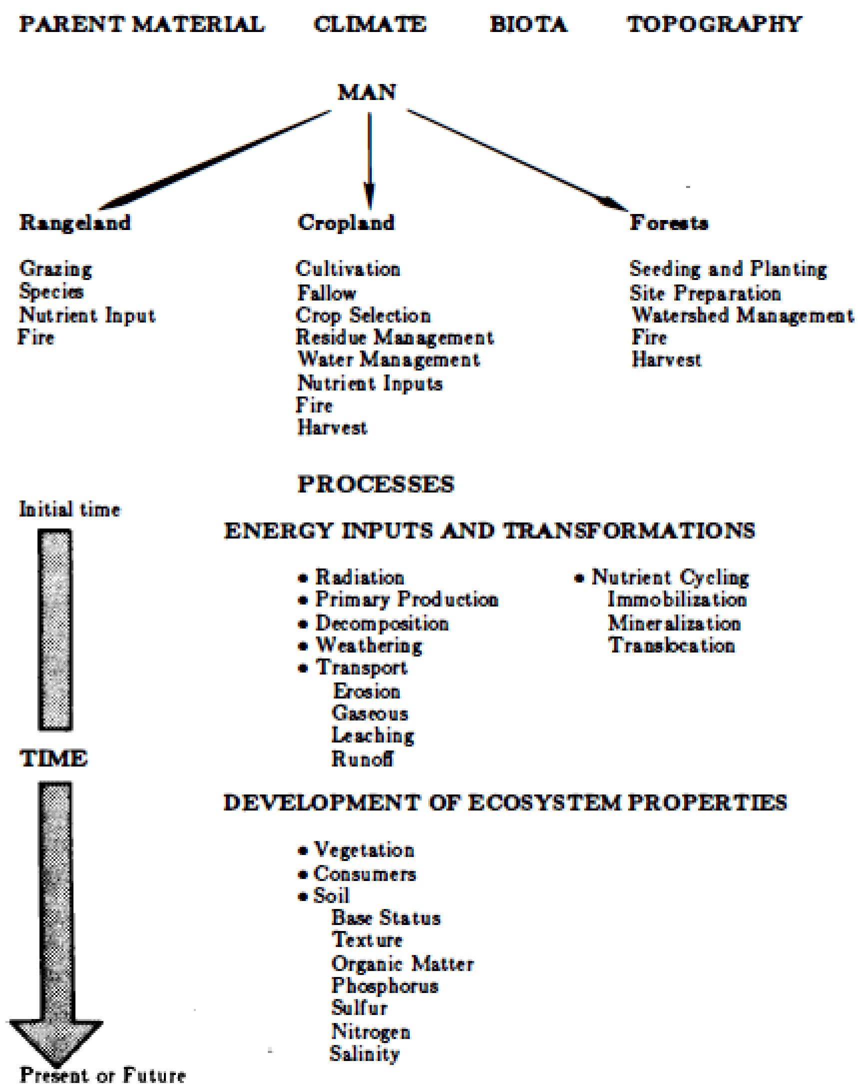

Fig 2: Relationships among driving variables, processes, and ecosystem properties [1]

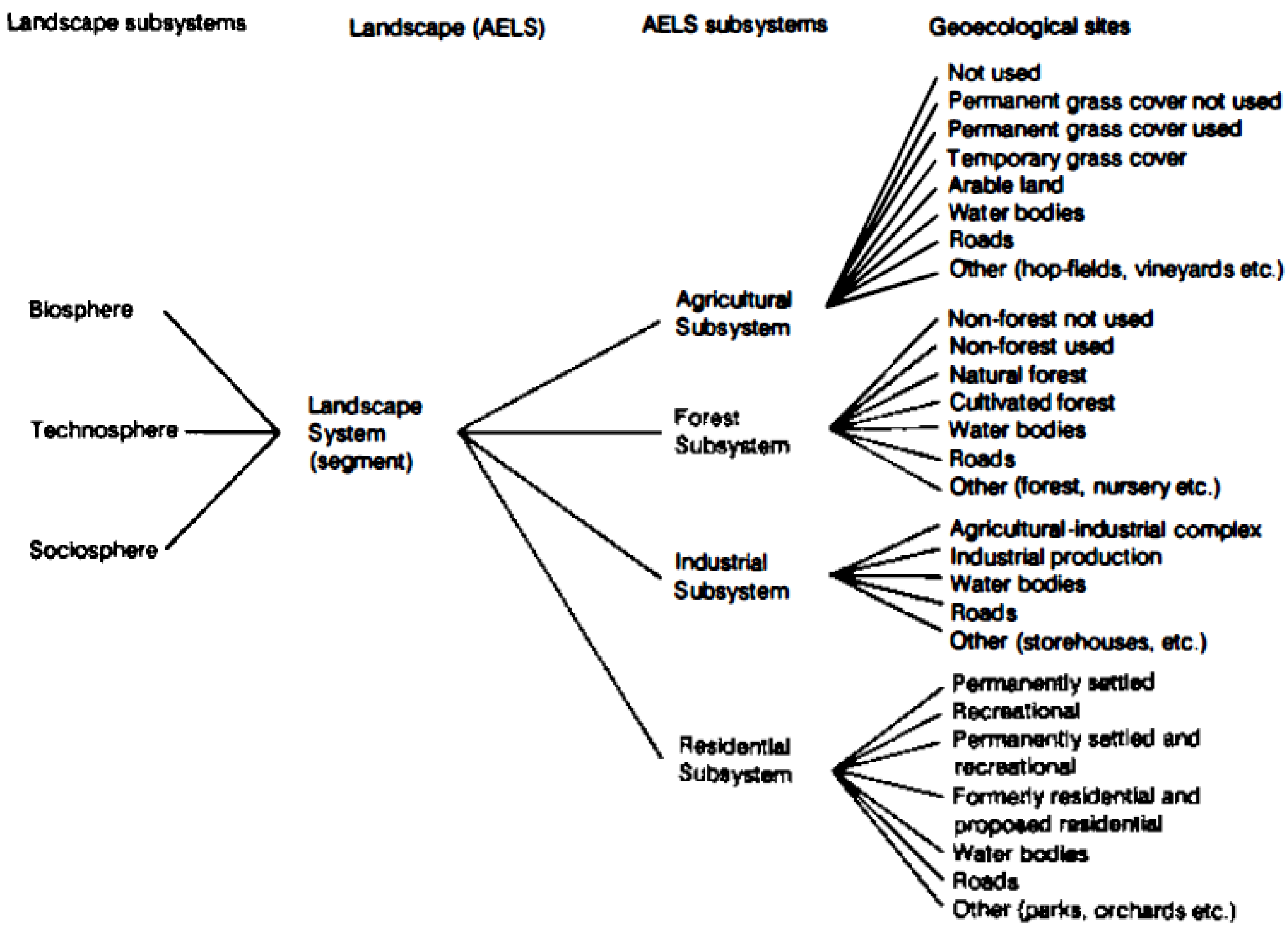

Fig 3: Anthropoecological landscape system structure

LIDAR

Classical Lidar devices are used to profile aerosol distributions in the lower atmosphere. Measurements are made by observing the relative backscattering of the intense, extremely short pulse of laser light as it interacts with suspended particles and molecules

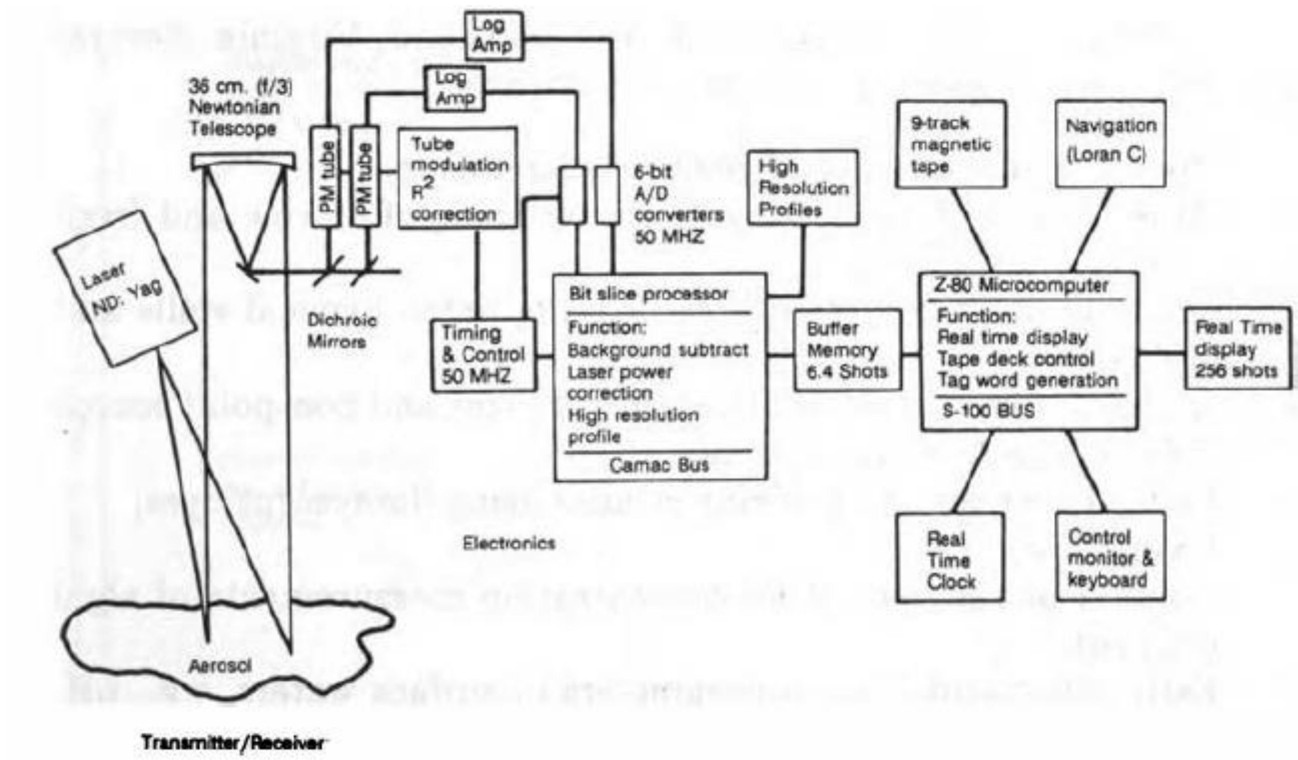

Fig 4: Airborne Lidar schematic [1]

GIS

A primary characteristic of monitoring networks and particularly remote sensing systems is that they produce a vast amount of spatially related data. Effective utilization of these large data sets depends on efficient geographic information systems (GIS) to process and convert the information into a usable form. Below are few of the parameters and features that can be used:

Flora and Fauna
Threatened/Endangered Species
Soil Morphology
Moisture Content
Aquifer Extent
Zone of Recharge
Hydraulic Conductivity
Groundwater Quality
- **Dissolved Organics**
- **pH**
- **Salinity**
- **Total Dissolved Solids**

Biological Activity (Soils)
Hydrology
Topography
Slope/Aspect/Elevation
Flood Prone Zones
Depth to Bedrock
Geographic Features

Critical Habitats
Soil Porosity
Mineral Composition
Texture
Depth to Aquifer
Flow Rate/Direction
Permeability
Meteorology/Climatology
- **Precipitation Record**
- **Temperature**
- **Wind Speed/Direction**
- **Temperature Gradient**

Soil Chemistry
- **Adsorption Coefficients**
- **Exchange Capacity**
- **Ion Speciation**
- **Mineral Content**
- **Leachability**
- **Solubility**

Fig 5: Various types of soils for GIS mapping [1]

Applied Research Team structure

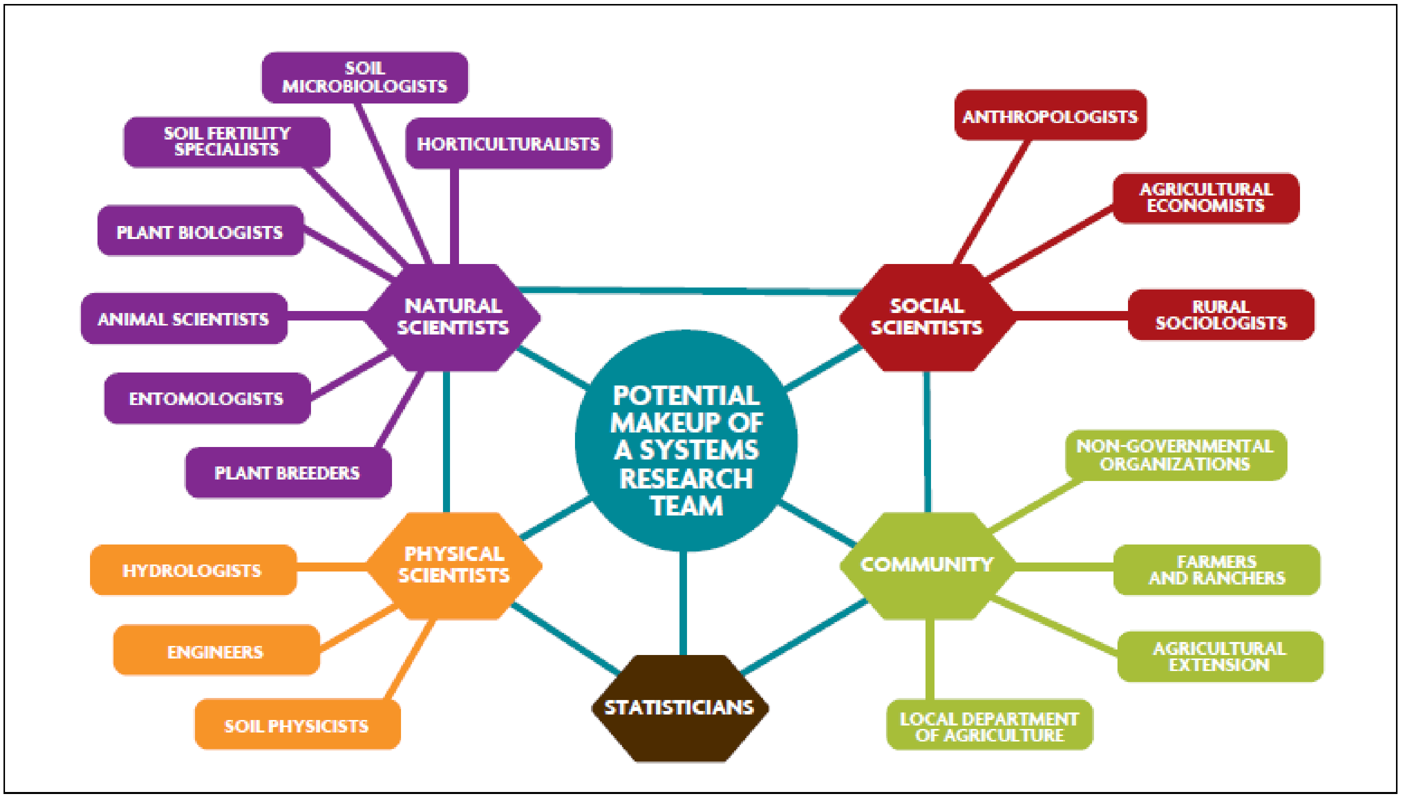

Fig 6: Applied Research Team sample structure [2], [3], [4]

Applied Research process:

This section provides the various steps involved in the domain specific experiments, the deviations taken, variables affected, systems and software used for business and data analysis, improvements observed from the existing process, product or a service.

Finding/ Discussions:

This section tabulates the key findings based on the experiments, interviews and processes taken. These findings can open up various hidden merits and weaknesses of the existing system or a product. A general recommendation based on the observations can be provided that can help converting this as a merit based quality driven product upgrade.

Conclusion:

This section provides the concluding remarks to the management that can further elaborate on the findings.

References:

[1]: http://www.fao.org/3/w7501e/w7501e04.htm
[2]: https://www.genomecanada.ca/en/programs/large-scale-science/funding-opportunities/large-scale-research-project-competitions/2018
[3]: https://www.agr.gc.ca/eng/agricultural-programs-and-services/agriscience-program-projects/applicant-guide/?id=1516993365431
[4]: https://www.ontariogenomics.ca/funding-opportunities/awarded-projects/large-scale-applied-research-project-competition-lsarp/